

SSPA Special - Application Questions

Sprint for Full Score

T3 reverse towards question + T7 text question · Two application trap total attack · 75 minutes · 1-to-3 Online Lesson

Corresponding textbooks: Scored test paper 1 + paper 2 application questions (accounting for about 40-50% of the paper) · Application of fractions · Application of equations · Reverse reasoning

Core traps: □ T3 reverse questions (reverse logic) · T7 application questions (question review + column expression + answer sentence) — ● SSPA compulsory exam

SSPA related: ● High-frequency Application questions in Paper 2 are the key to determine the score grade (A/B/C grade watershed)

Prerequisite knowledge: L10 fraction application questions · L14–L15 Algebra and equations · L19 Simulation 2 Inverse Questions

The goal of this class: ❶ Zero mistakes in fraction word problems ❷ Quick expression of equation word problems ❸ Automation of reverse logic in reverse questions ❹ Complete sentence answering habits

Student Name:

Class:

Date:

Time Spent:

I. Application question warm-up (total 4 questions, 5 minutes)

#	Question	Difficulty	Working Space (must write column + answer sentence)
W1	Xiao Ming had 12 candies and ate $\frac{1}{3}$. How many pills did you eat? $\frac{1}{3}$. How many pills did you eat?		
W2	There are 24 egg tarts in a box. My sister ate $\frac{1}{4}$, how many are left? $\frac{1}{4}$, how many are left?		
W3	Adding 8 to a number equals 25. What is that number? (Use the equation: let this number be x)		
W4	A box of apples, 15 of which are sold, leaves 28 apples. How many were there? (Use backwards reasoning)		

II. Core Knowledge + Worked Examples

Knowledge point 1: Trap of fraction word questions (T7 text questions) ●

SSPA required test

Four-step method for solving fraction application problems:

- ① **Circle keywords** $\frac{1}{3}$: Circle "total", "total" (addition), "remainder", "remaining" (subtraction), "how many times of..." (multiplication) Confirm "whole" $\frac{1}{3}$: The fraction in the question is accounted for
- ② **What** What fraction of? (The whole $\neq$ must be the total number) **Column calculation** $\frac{1}{3}$: **Fraction $\times$ whole amount = partial amount (multiply first and then reduce)** Complete answer sentence $\frac{1}{3}$: Must write "Answer:... (Quantity) (Unit)" - Do not write an answer sentence = 1-2 points will be deducted for each step.
- ③ **Column calculation**: Fraction $\times$ whole quantity = partial quantity (multiply first and then reduce)
- ④ **Complete answer**: Must write "Answer:... (Quantity) (Unit)" - Do not write an answer sentence = 1-2 points will be deducted for each step.

Example question KP1-1: Find partial quantities of fractions (T7 classic questions)

There are 30 chocolates in a box. My brother ate a box of $\frac{1}{3}$. How many pills did he eat? My sister ate the remaining $\frac{2}{5}$ Box, how many pills did you eat? My sister ate the rest $\frac{1}{3}$, how many pills did my sister take?

 **Common mistakes**

 **Correct solution**

Sister: $30 \times \frac{1}{3} = 10$

"The remaining 13" is the "remaining 18 grains" as a whole, not the original 30!

Brother: $30 \times \frac{2}{5} = 12$ capsules

Remainder: $30 - 12 = 18$ capsules

Sister: $18 \times \frac{1}{3} = 6$ capsules

"The rest" → The new whole must be calculated first!

Example questions KP1-2: Fraction comparison problems

Rope A is $2\frac{1}{2}$ meters long, and rope B is longer than A $\frac{3}{4}$ meters. How many meters long are the two ropes?

✗ Common mistakes

$$2\frac{1}{2} \times \frac{3}{4} = ?$$

"Longer than..." is addition, not multiplication!

✓ Correct solution

$$B = 2\frac{1}{2} + \frac{3}{4} = 3\frac{1}{4}$$

$$A+B = 2\frac{1}{2} + 3\frac{1}{4} = 5\frac{3}{4} \text{ rice}$$

First find the length of B (addition), then add A to find the sum.

 **Tips: "Circle the key words, determine the overall quantity, formulate it step by step, and write the answer sentence"**

 The most frequent error in T7: "The remaining fraction" - the whole has changed and the original total can no longer be used!

KP1 synchronization practice — fractionapplicationquestion (question 1-14)

#	Question	Difficulty	Working Space (column → calculate → answer sentence)
1	There are 40 cookies in a box. Beautifully ate $\frac{3}{8}$ box. How many pieces did she eat?		
2	A bottle of juice contains $\frac{3}{4}$ liters. Drink $\frac{1}{3}$ liters. How many liters are left?		
3	SEP in the garden grows roses, $\frac{2}{5}$ grows chrysanthemums. What percentage of the garden is occupied by flowers? $\frac{1}{4}$ Plant chrysanthemums. What percentage of the garden is occupied by flowers?		

4	SEP in the granary is for large rice, $\frac{1}{6}$ is for brown rice. What percentage are the two together? What fraction is left? $\frac{2}{5}$ Put brown rice. What percentage are the two together? What fraction is left?		
5	Xiao Ming has \$120. Use $\frac{1}{4}$ to buy books, and then use $\frac{1}{3}$ to buy stationery. How many yuan did he spend in total?		
6	The original water bottle contains $\frac{3}{4}$ liters, then pour $\frac{1}{3}$ liters into it. How many liters are there now? (Answer with mixed fractions)		
7	A rope is $2\frac{1}{2}$ meters long. Used $\frac{3}{4}$ meters. How many meters are left? (Answer with fractions)		
8	24 hours a day. Xiao Ming uses $\frac{1}{3}$ to sleep and $\frac{1}{4}$ to go to school. How many hours are spent sleeping and going to school?		
9	There are 36 candies in a box. The younger brother ate SEP , and the younger sister ate the remaining SEP . How many pills did your sister take? $\frac{1}{4}$, my sister ate the rest $\frac{1}{3}$. How many pills did your sister take?		
10	The cake shop sold $\frac{2}{5}$ cakes in the morning and $\frac{1}{3}$ cakes in the afternoon. How many percent were sold in total throughout the day? If 60 units were made throughout the day, how many units were sold?		
11	A section of railway, the first section is $\frac{3}{5}$ kilometers long, and the second section is longer than the first section $\frac{1}{4}$ kilometers. How many kilometers are the two sections in total?		
12	A packet of sugar weighs $\frac{3}{4}$ kg. Used to make $\frac{1}{2}$ kilo cake. What fraction of the original sugar is left? How many kilograms are left?		
13	There are 40 people in the class. Those who failed $\frac{3}{10}$ in the math test, and those who passed $\frac{2}{7}$ got an A grade. How many people are in Class A? (Note that the two "wholes" are different!)		
14	The school garden covers an area of SEP . The garden $\frac{1}{5}$ is planted with flowers and the rest is grass. What percentage of the school's total area is grassland? $\frac{3}{4}$ Plant flowers and the rest is grass. What percentage of the school's total area is grassland?		

Knowledge point 2: Equation word problem trap (T3 inverse + T7 text questions) ● SSPA required test

Solutions to equation word problems:

- ① **Let the unknown number** |||SEP|||: Find the "required quantity" → set it to x (or other letters) Find the equivalence relationship |||SEP|||: "equal to", "total" and "remaining" in the question → This is the equal sign
- ② **Column equations** |||SEP|||: Use x to express the equivalence relationship → Write out the equation Solve the equation |||SEP|||: Shift terms (reciprocal addition and subtraction, reciprocal multiplication and division) → Find x
- ③ **Verify + answer sentence** |||SEP|||: Substitute the original question to check whether it is reasonable → Write an answer sentence: Use x to express the equivalent relationship → write the equation
- ④ **Solve equations**: Shift terms (reciprocal addition and subtraction, reciprocal multiplication and division) → find x
- ⑤ **Verify + answer**: Return to the original question to check whether it is reasonable → Write an answer sentence

①

Let x

The unknown quantity is equal to the required quantity

②

Find equal amounts

"Equal" and "shared" are equal signs

③

series of equations

Write the equation relationship in terms of x

④

solution + answer

Move items to solve and write complete sentences

Example KP2-1: Standard equation word problems

Xiao Ming has some stickers. After he gave his sister 15, there were 28 left. How many were there originally?

✗ **Error**

$$28 - 15 = 13$$

Forward subtraction - but the original number should be more than the remaining number

✓ **Correct**

Assume that there are x original

$$x - 15 = 28$$

$$x = 28 + 15 = 43$$

Given = subtracted, the last remaining 28 → the original must be > 28

Example KP2-2: Inverse fractional equation

A barrel of oil, after using $\frac{2}{5}$, there are 12 liters left. How many liters did it originally have? After that, there are 12 liters left. How many liters did it originally have?

✗ Error

$$12 \times \frac{2}{5} = 4.8 \text{ liters}$$

25 is used, the remaining 12 liters are 35 instead of 25

✓ Correct

Assume that the original x liter

$$\text{Used } \frac{2}{5}x \rightarrow \text{leftover } \frac{3}{5}x$$

$$\frac{3}{5}x = 12 \rightarrow x = 20 \text{ liters}$$

$$\text{Remaining} = 1 - \text{Used} = 35$$

⚠ The key to T3 reverse question: Ask yourself "What fraction of what is known?" - used 25 → the remaining is 35 (not 25)

KP2 synchronization practice — equation and inverse toward applicationsquestion (question 15-30)

#	Question	Difficulty	Working Space (let unknown → equation → solution → answer sentence)
15	3 times a number plus 5 equals 26. Find this number. (let this number be x)		
16	Xiao Ming saved some money. Mom gives him another \$45, so he now has \$120. How many yuan was it originally?		
17	A box of apples, after selling 28, 35 are left. How many apples were there?		
18	Subtract 12 from a number and then divide it by 5. The result is 8. Find a certain number.		
19	Four times a number minus 7 equals 29. Find this number.		
20	A barrel of oil, after using 8 liters, is left with the original $\frac{3}{5}$. How many liters did it originally have?		
21	Xiao Ming used the saved $\frac{1}{4}$ to buy books, and then used \$30 to buy stationery, a total of \$70 for the two times. How many yuan did you originally save?		
22	The sister is 5 years older than the brother. The sum of their ages is 25. How old is your brother?		
23	The length of a rectangle is 3 times its breadth and its perimeter is 48 cm. Seek length and breadth.		
24	Xiao Ming divided some candies into 4 equal parts, and after subtracting 2 pieces from each piece, there were 5		

	pieces. How many candies were there?		
25	A bottle of juice, I drank $\frac{1}{3}$ on the first day, and drank the remaining $\frac{1}{2}$ on the second day, leaving $\frac{1}{2}$ liters. How many liters did it originally have?		
26	The number of stickers Xiaomei has is twice that of Xiaohua, but 3 less. Xiaohua has 20 pictures. How many pictures does Xiaomei have?		
27	Add 12 to a number, multiply it by 3, and subtract 8. The result is 64. Find a certain number.		
28	A bunch of books, $\frac{1}{5}$ are Chinese books, and the rest are English books. There are 24 more English books than Chinese books. How many books are there in total?		
29	The father is 45 years old and the son is 12 years old. In how many years will the father's age be twice that of his son?		
30	A water tank is filled with $\frac{1}{4}$ water, and then 200 liters are released, and the remaining water is half of the original amount. How many liters did it originally have?		

Knowledge point 3: Comprehensive application of strategies - question review = half of the question solved ● SSPA

Universal problem-solving framework for word problems (6 steps):

① **Read the question** |||SEP|||: Read it twice completely, without skipping lines Circle keywords |||SEP|||:

Numbers, units, action words (buy/sell/use/left/total/ratio/times)

② **Drawing/list** |||SEP|||: Use stick figures and tables to organize information for complex

questions Judgment type |||SEP|||: Forward (inferring the unknown from the known) or reverse (inferring backward from the result)?

③ **Select Tool** |||SEP|||: Arithmetic (step-by-step equation) or algebra (let x be an equation)? Answer Check

|||SEP|||: Do you have an organization? Is it reasonable? Is it right to take it back?

④ **Judgment type**: Forward (inferring from the known to the unknown) or backward (inferring from the results backwards)?

⑤ **Select tool**: Arithmetic (step-by-step formula) or algebra (let x be a series of equations)?

⑥ **answer check**: Do you have a unit? Is it reasonable? Is it right to take it back?

Example KP3-1: Multi-step comprehensive application questions

The store has 120 apples. Sell $\frac{1}{3}$ in the morning and sell the remaining $\frac{2}{5}$ in the afternoon. How many were sold in total throughout the day? How many are left?

How many were sold in total throughout the day? How many are left?

✗ Error

$$120 \times \frac{1}{3} + 120 \times \frac{2}{5}$$

The 25 in the afternoon is the remainder, not the 120!

✓ Correct

AM: $120 \times \frac{1}{3} = 40$ pieces

Remainder: 80

Afternoon: $80 \times \frac{2}{5} = 32$

Total sold: 72 pieces, remaining: 48 pieces

Every step must confirm what the "whole" is

Example KP3-2: Equations vs Arithmetic Choices

A sum of money, after using $\frac{3}{8}$, and then using \$45, there is still \$75 left. How many yuan was it originally? Finally, I spent another \$45, leaving \$75 left. How many yuan was it originally?

✗ Arithmetic is error-prone

$$\$75 + \$45 = \$120 \rightarrow \text{original} = ?$$

\$120 is what is left after using \$38 and then \$45. You have to work backwards two steps.

✓ Equation method is clearer

Assume the original \$x

$$x - \frac{3}{8}x - 45 = 75$$

$$\frac{5}{8}x = 120$$

$$x = 192$$

The equation expresses all relationships at once, making it difficult to miss

⚠ The ultimate formula for application questions: "The fraction must be determined as a whole, the reverse must be reversed, the units must be consistent, and the answer must be complete."

KP3 comprehensive application Sprint (question 31-43)

#	Question	Difficulty	Working Space
31	The bookstore has 240 books. Novels account for $\frac{1}{4}$, popular science $\frac{1}{3}$, the rest are comics. How many comics are there?		

32	Xiaomei has some stickers. After giving 18 pieces to a friend, divide the remaining pieces into 5 equal parts, each with 7 pieces. How many were there?		
33	A number subtracted from $\frac{1}{3}$ and then multiplied by 6, the result is 24. Find this number.		
34	Juice 1.5 liters. How many cups can be poured using 0.25 liters per cup? Pour the rest into the $\frac{1}{4}$ liter cup. How much is left before it is full?		
35	Divide \$360 between A, B, and C. A gets $\frac{1}{3}$, B gets the remaining $\frac{1}{2}$, and C gets the balance. How much does C get?		
36	A water tank originally contains several liters of water. After withdrawing $\frac{1}{5}$, add another 30 liters, now you have 90 liters. How many liters did it originally have?		
37	A has \$x and B has \$(x+15). A total of \$95 for two people. x = ?		
38	After cutting off $\frac{2}{7}$ from a rope, then cut off 5 meters, leaving 12 meters. How many meters long was the rope originally?		
39	The average score in Chinese, English and Mathematics is 85 points. Known Chinese is 88 points and English is 82 points. What's the score in math?		
40	Each pack of sugar weighs $\frac{3}{4}$ kg. There are $5\frac{1}{4}$ kilograms of sugar. How many bags can be filled? How many kilograms are left?		
41	This year, my mother is 4 times older than Xiao Ming. Five years later, my mother will be three times as old as Xiao Ming. How old is Xiao Ming this year?		
42	A barrel of oil is 20 liters. The whole bucket of $\frac{1}{4}$ was used on the first day, the remaining $\frac{2}{5}$ from the first day was used on the second day, and the remaining $\frac{1}{3}$ was used on the third day. How many liters are left?		
43	There are boys and girls in class one. Boys account for $\frac{3}{5}$, with 6 more boys than girls. How many people are there in the class? (Tip: Girls account for $\frac{2}{5}$, extra $\frac{1}{5}$ corresponds to 6 people)		

III. application question add points special training (question 44-47 · No. 1 pick)

#	Question	Difficulty	Working Space
44	An item is first increased in price $\frac{1}{5}$, and then reduced in price $\frac{1}{5}$, resulting in a selling price of \$192. What was the original selling price? (Note: The overall difference between the two $\frac{1}{5}$ is different!) $\frac{1}{5}$, resulting in a selling price of \$192. What was the original selling price? (Note: twice $\frac{1}{5}$ The overall difference!)		
45	A, B, and C share a sum of money. A gets $\frac{1}{3}$, B gets $\frac{1}{4}$, and C gets \$250. How much is the total amount of money?		
46	A pool can be filled with water in $\frac{1}{3}$ hours and can be emptied in $\frac{1}{2}$ hours. If water is filled and drained at the same time, how many hours will it take for it to be full? (Note: Net injection rate = $\frac{1}{3} - \frac{1}{2}$? Negative number? - This question requires thinking about what situations will it be full) $\frac{1}{3} - \frac{1}{2}$? negative number? —This question requires thinking about what situations will be satisfied)		
47	A fraction whose numerator and denominator sum is 52. If 3 is added to the numerator and 1 is subtracted from the denominator, the fraction is equal to 1. Find the original fraction.		

IV. The Lesson core common error summary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)☐ Explain the lesson formula to classmates☒ 本堂自我檢查 (完成後打剔)☐ 我識得分辨每個知識點嘅陷阱☐ 我能夠獨立完成 🌿 基礎題☐ 我能夠挑戰 🌿 進階題☐ 我記得住口訣

#	common error	trap	Correct Approach
1	"The remaining fraction" is overall wrong	T7	Every time "remaining" appears → recalculate the new whole
2	"More/longer than..." Misuse of multiplication	T7	"There are more B than A" = A + B (it's addition!)
3	The reverse direction of the reverse question is wrong	T3	Work backwards from the results: addition, subtraction, multiplication, division, reversal of order
4	When applying fractions, forget to determine the whole	T7	For each fraction one must ask "What fraction of what is this?"
5	The equation is written directly without any unknowns.	T3	You must first write "let... be x" and then formulate the equation
6	Missing an answer sentence / Missing a unit	T7	The last line of each question must be written "Answer:..."
7	used/remaining confusion	T3	Used $\frac{1}{4}$ → left $\frac{3}{4}$
8	Skip steps and miss steps in multi-step questions	T7	Write it in steps, labeling each step "Step 1:...Step 2:..."

Strategies for perfect scores on applied questions 🟡 SSPA top scorer

Three tips to win the application questions in Paper 2:

- ① **Answer step by step |||SEP|||**: Don't give the answer in one step! Write each step clearly → Even if the answer is wrong, you can still get step points
Check the calculation and substitution |||SEP|||: Substitute the answer into the original question to check → Correct it immediately if it is unreasonable
- ② **Easy first, then difficult |||SEP|||**: Do the questions you are confident about first to ensure you get the basic points, and then attack the difficult questions: Substitute the answer into the original question to check → Correct if unreasonable immediately
- ③ **Easy first, then difficult**: Do the questions you are confident about first to ensure you get the basic points, and then attack the difficult questions.

🧠 **Tips for application questions: "Circle keywords, determine whole quantities, list step-by-step formulas, attack and inverse equations, and write answer sentences."**

今日學咗咩？小朋友學咗「SSPA Special - Application Questions Sprint for Full Score」。

最易錯嘅 3 個陷阱：留意老師圈出嘅陷阱題目

你可以問小朋友：「你可唔可以解釋「SSPA Special - Application Questions Sprint for Full Score」嘅最重要口訣俾我聽？」

溫馨提示：唔需要識教數學 — 只需要問小朋友「點解咁計？」同「有冇檢查陷阱？」就夠。

 教師參考：熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號： 挑戰層 17-21